

Uncertainty and Sentiment in ARK Company Filings

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Repository: github.com/victor4628/FRE7871-NLP

I examine financial negative language and uncertainty in 1,624 eligible 10-K/10-Q filings from 91 ARK-held issuers, filed during 2021-2025. The clearest evidence is a rise in negative language within annual-report issuers. Uncertainty trends depend on report type and weighting. Its association with subsequent volatility weakens after controlling for existing volatility. Four-day return estimates are imprecise; they do not establish either predictability or the absence of an effect.

Table 1 Sample construction

Security and issuer screen	Removed	Remain
Unique cleaned securities in frozen snapshot	0	124
SEC ticker-to-CIK mapping found	7	117
At least one 2021-2025 10-K/10-Q	24	93
Distinct issuers after multi-class consolidation	1	92

Unique filing screen	Removed	Remain
Downloaded security-filing records	0	1,702
Duplicate Alphabet records (retain GOOGL)	20	1,682
Valid acceptance timestamp	0	1682
At least 2,000 narrative words	0	1682
Exclude explicitly identified shell companies	58	1624

Market screen (separate branches)	Vol: removed / left	Return: removed / left
Starting unique filings	0 / 1624	0 / 1624
Complete 63-day pre-filing returns and positive volatility	32 / 1592	32 / 1592
Dated instantaneous common shares and positive market value	0 / 1592	0 / 1592
Complete outcome window and benchmark	1 / 1591	0 / 1592

The initial 31 losses are 7 unresolved SEC ticker mappings and 24 mapped securities without an eligible 10-K/10-Q in the window. Filing counts for those excluded securities are not observed and are not invented. The 93 eligible securities represent 92 issuers because GOOG and GOOGL share filings. Explicitly reported shells are excluded. The optional \$3 price cutoff was removed at the student's request; low-priced stocks remain eligible. Legacy shell checkboxes and ordinary-share cover wording were corrected. All download and parsing failures were zero.

Measures and identification

The instructor-specified LM lists contain 2,355 negative words and 297 uncertainty words. They overlap on 40 words, so the concepts are distinct but their empirical measures are not independent. Ten nonzero Negative flags actually mark removed entries; the required list is retained for comparability, with an active-only list as a sensitivity.

For term i in document j , I implement equation (1) as:

$w_{ij} = [(1 + \ln tf_{ij}) / (1 + \ln a_j)] \ln(N / df_i)$ for $tf > 0$; otherwise zero.

Here a_j is total tokens divided by distinct tokens, not total document length. A category score sums its term weights. Proportions divide category occurrences by all tokens. IDF is recomputed on the exact document corpus used by each regression, including restricted sensitivities. This is full-corpus, retrospective scoring, not an out-of-sample forecast.

Parsing retains visible inline-XBRL text and removes hidden resources and mostly numeric tables (digit share above 15%). The corrected parser retains 62,590,430 tokens across unique downloads versus 48,537,683 under the starter parser, which also deleted visible tagged narrative. The starter uppercase tokenizer is retained; there is no stemming or stopword removal.

Table 2 Summary statistics by form

Form / measure	N	Mean	SD	P25	Median	P75
10-K / Negative %	396	2.347	0.406	2.084	2.361	2.620
10-K / Negative TF-IDF	396	163.684	54.480	122.192	155.045	204.915
10-K / Uncertainty %	396	1.947	0.240	1.803	1.969	2.119
10-K / Uncertainty TF-IDF	396	26.643	8.120	21.062	25.542	31.201
10-Q / Negative %	1,228	2.133	0.948	1.285	1.952	3.020
10-Q / Negative TF-IDF	1,228	80.297	66.244	24.021	56.244	131.109
10-Q / Uncertainty %	1,228	1.829	0.587	1.327	1.669	2.419
10-Q / Uncertainty TF-IDF	1,228	13.505	7.271	7.788	11.909	18.719

Negative-uncertainty correlations: 10-K prop: 0.722; 10-K tfidf: 0.838; 10-Q prop: 0.851; 10-Q tfidf: 0.921. Shared risk topics, overlapping vocabulary and document structure can make the scores move together. The separate outcome tests do not assume statistical independence.

Event timing and controls

Day 0 is the first exchange session closing strictly after the SEC acceptance timestamp, including early closes; 921 core filings move from their filing date. Four-day return is stock buy-and-hold return over [0,3] minus SPY buy-and-hold return. Volatility is daily-return SD times sqrt(252): [-63,-1] before and [4,66] after, excluding the four-day reaction. All 63 returns are required; missing prices are never filled.

Size is log market value: dated common shares from the same filing times day -1 nominal price, with split units aligned. It is an optional research control, included because company scale may affect both disclosure language and market behavior. Multiple common classes are summed and valued at the selected class price, an approximation. Weighted-average EPS shares are excluded. Tables 5-6 compare no controls, size only, prior volatility only and both on a common sample. The primary models contain no other controls or fixed effects. Prior return and turnover are not calculated. The 63 preceding returns provide prior volatility; size needs one preceding closing price and a share count. Standard errors cluster by issuer.

Table 3 Words driving each measure

Shares below divide each word count by all occurrences of words on its own list, not by all words in the filings. The top 30 words account for 46.4% of Negative occurrences and 90.4% of Uncertainty occurrences. Uncertainty is particularly concentrated, which makes common-word downweighting consequential.

Rank	Negative word	List share	Uncertainty word	List share
1	LOSS	5.12%	MAY	35.22%
2	ADVERSELY	4.13%	COULD	18.54%
3	LOSSES	3.11%	RISK	4.98%
4	CLAIMS	2.97%	RISKS	4.48%
5	ADVERSE	2.77%	BELIEVE	2.93%
6	AGAINST	2.38%	APPROXIMATELY	2.56%
7	UNABLE	1.96%	ASSUMPTIONS	1.92%
8	LITIGATION	1.95%	INTANGIBLE	1.64%
9	HARM	1.87%	POSSIBLE	1.26%
10	FAILURE	1.81%	UNCERTAINTIES	1.16%
11	FAIL	1.34%	FLUCTUATIONS	1.12%
12	IMPAIRMENT	1.24%	MIGHT	1.06%
13	NEGATIVELY	1.14%	UNCERTAIN	1.02%
14	PENALTIES	1.14%	ANTICIPATED	0.99%
15	DIFFICULT	1.14%	PREDICT	0.95%
16	NEGATIVE	1.02%	VOLATILITY	0.94%
17	CRITICAL	1.01%	DEPEND	0.93%
18	DELAYS	0.94%	UNCERTAINTY	0.89%
19	DECLINE	0.92%	DIFFER	0.85%
20	DAMAGES	0.91%	ANTICIPATE	0.80%
21	RESTATED	0.85%	CONTINGENT	0.77%
22	LIMITATIONS	0.85%	EXPOSURE	0.77%
23	DELAY	0.84%	VARIABLE	0.69%
24	VOLATILITY	0.77%	DEPENDS	0.68%
25	CHALLENGES	0.76%	PENDING	0.65%
26	FINES	0.72%	DEPENDENT	0.58%
27	CLOSING	0.69%	CONTINGENCIES	0.55%
28	HARMED	0.69%	PROBABLE	0.51%
29	DISRUPTIONS	0.69%	ASSUMED	0.51%
30	INVESTIGATIONS	0.66%	VARY	0.48%

A frequently used word can dominate proportional counts yet convey little cross-document distinction. Equation (1) reduces that contribution through document frequency, while log term frequency dampens repetition. A high TF-IDF value can also reflect unusual vocabulary and document structure; it is not a probability of bad news.

Figure 1 Tone and market uncertainty

Each line first averages filings within issuer, form and filing quarter, then weights issuers equally. Annual and quarterly reports remain separate. VIX is the quarterly mean on the right axis; it is a market comparison, not the dependent variable in the firm-volatility regressions.

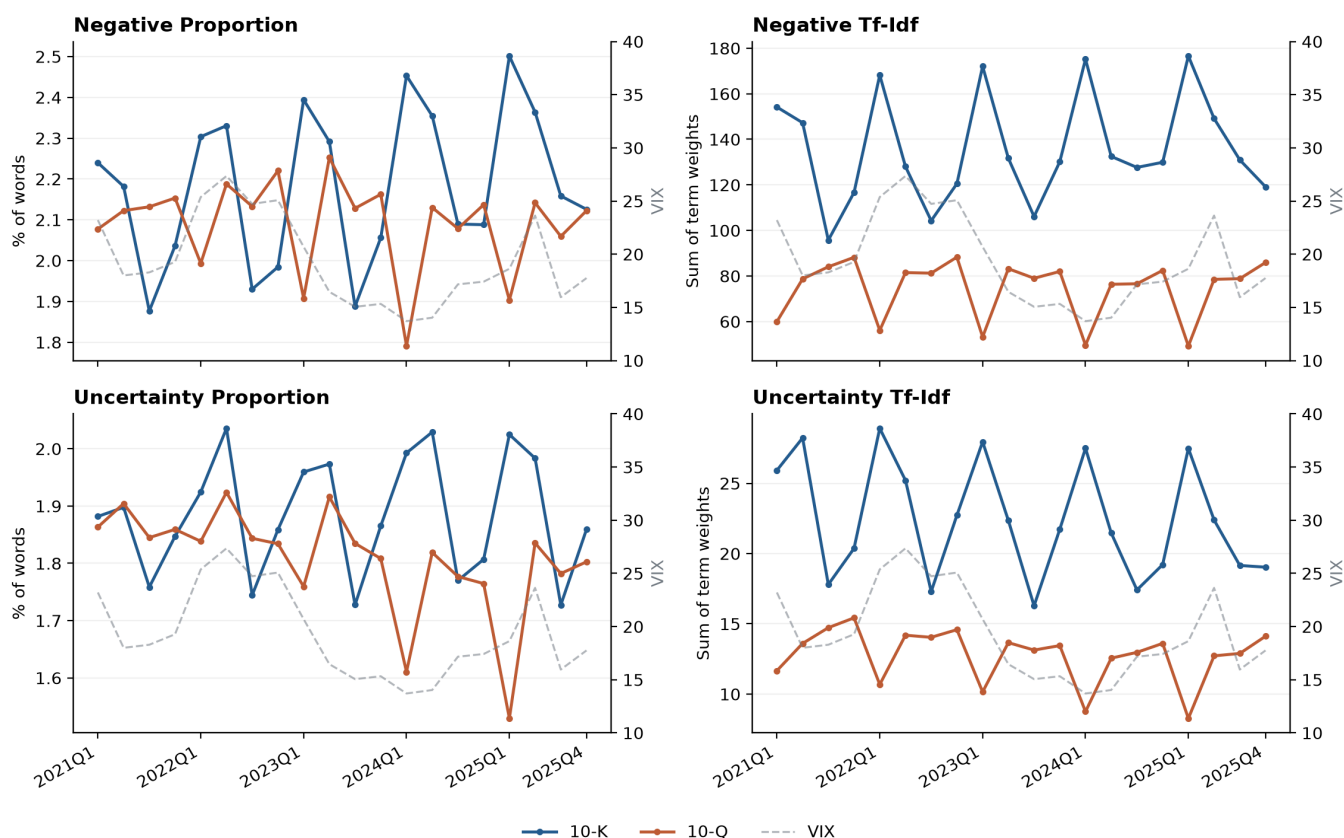


Table 4 Aggregate and within-issuer trends

Form	Measure	Agg. slope	OLS t	NW t	Within slope	p	Holm p
10-K	Negative %	0.0318	6.53	9.50	0.0345	<0.001	<0.001
10-K	Negative TF-IDF	0.0176	2.85	4.73	0.0133	0.002	0.007
10-K	Uncertainty %	0.0119	1.83	1.73	0.0249	<0.001	<0.001
10-K	Uncertainty TF-IDF	-0.0124	-1.44	-2.23	-0.0077	0.077	0.154
10-Q	Negative %	-0.0060	-2.57	-3.35	-0.0042	0.332	0.332
10-Q	Negative TF-IDF	-0.0062	-4.16	-4.06	-0.0072	0.113	0.154
10-Q	Uncertainty %	-0.0161	-4.30	-4.52	-0.0109	0.019	0.038
10-Q	Uncertainty TF-IDF	-0.0179	-6.28	-6.16	-0.0181	<0.001	0.001

Slopes are SDs per quarter. Aggregate models include quarter-of-year effects, with Bartlett Newey-West SEs (4 lags, finite-sample correction) alongside naive OLS t statistics. Within-issuer models include issuer and seasonal effects and issuer-clustered SEs; issuers observed in only one quarter are omitted from that test. Holm adjustment covers the four within-issuer tests separately for each weighting scheme. Redundant fixed-effect columns are removed algebraically.

The within-issuer results, rather than the pooled chart, support the main trend interpretation. Annual-report negative language rises under both weights. Uncertainty does not have one universal trend: annual-report proportional uncertainty rises while its TF-IDF counterpart does not; quarterly-report uncertainty declines. Changing report mix and issuer mix therefore cannot be ignored. With only 20 aggregate quarters, even corrected aggregate inference is fragile.

Table 5 Uncertainty and subsequent volatility

The outcome is log annualized volatility on days [4,66]. Each panel compares exactly the same filings and standardized tone under four control choices. None means tone plus intercept only; size is log market value and volatility is log prior volatility. Parentheses contain issuer-clustered SEs.

Proportion

Regressor / statistic	None	Size only	Volatility only	Both
Tone (1 SD)	0.1204 (0.0368)	0.1170 (0.0261)	0.0257 (0.0102)	0.0381 (0.0089)
Log market value	-	-0.1237 (0.0131)	-	-0.0429 (0.0069)
Log prior volatility	-	-	0.7487 (0.0269)	0.6417 (0.0367)
N	1,591	1,591	1,591	1,591
Tone p	0.002	<0.001	0.014	<0.001
Tone Holm p	NA	NA	NA	<0.001

TF-IDF

Regressor / statistic	None	Size only	Volatility only	Both
Tone (1 SD)	0.1382 (0.0267)	0.1064 (0.0234)	0.0418 (0.0096)	0.0435 (0.0092)
Log market value	-	-0.1188 (0.0138)	-	-0.0405 (0.0069)
Log prior volatility	-	-	0.7408 (0.0282)	0.6443 (0.0368)
N	1,591	1,591	1,591	1,591
Tone p	<0.001	<0.001	<0.001	<0.001
Tone Holm p	NA	NA	NA	<0.001

Proportion: adding prior volatility changes the tone coefficient from 0.1204 to 0.0257 without size, and from 0.1170 to 0.0381 holding size constant. TF-IDF: adding prior volatility changes the tone coefficient from 0.1382 to 0.0418 without size, and from 0.1064 to 0.0435 holding size constant.

Prior volatility accounts for much of the pooled association. With both controls, the remaining association is significant after Holm adjustment across the four both-control outcome tests. This is a conditional association, not evidence that words cause volatility. The other columns show sensitivity to the controls; their p values are unadjusted.

Sensitivity of the model with both controls

Specification	Proportion: coefficient / p	TF-IDF: coefficient / p
10-K	0.0566 / 0.009	0.0646 / <0.001
10-Q	0.0408 / <0.001	0.0417 / <0.001
firm FE	-0.0088 / 0.393	-0.0047 / 0.692
two way	0.0381 / <0.001	0.0435 / 0.046

Form-specific samples re-estimate IDF. The firm FE sensitivity adds issuer, calendar-quarter and form effects; both uncertainty coefficients then become negative and insignificant. A robust within-issuer predictive relationship is therefore not established. Two-way clustering has only 20 time clusters; invalid nuisance variances are logged in the notebook.

Table 6 Sentiment and four-day excess returns

The outcome is stock minus SPY buy-and-hold return over [0,3], in percentage points. The same four models are estimated on a common sample within each weight. Parentheses contain issuer-clustered SEs; a dash means the regressor is omitted, not estimated to be zero.

Proportion

Regressor / statistic	None	Size only	Volatility only	Both
Tone (1 SD)	-0.3546 (0.2808)	-0.3657 (0.2756)	-0.2614 (0.2848)	-0.2093 (0.2845)
Log market value	-	0.0498 (0.1172)	-	-0.1087 (0.1751)
Log prior volatility	-	-	-0.9208 (0.5999)	-1.1967 (0.8579)
N	1,592	1,592	1,592	1,592
Tone p	0.210	0.188	0.361	0.464
Tone Holm p	NA	NA	NA	0.484

TF-IDF

Regressor / statistic	None	Size only	Volatility only	Both
Tone (1 SD)	-0.5068 (0.2973)	-0.5043 (0.3063)	-0.3905 (0.3233)	-0.3781 (0.3209)
Log market value	-	0.0101 (0.1236)	-	-0.1258 (0.1742)
Log prior volatility	-	-	-0.8153 (0.6381)	-1.1191 (0.8644)
N	1,592	1,592	1,592	1,592
Tone p	0.092	0.103	0.230	0.242
Tone Holm p	NA	NA	NA	0.484

All four specifications produce negative but statistically imprecise tone coefficients. With both controls, the estimates are -0.209 and -0.378 percentage points per SD. Failure to reject zero does not establish the absence of an economically meaningful effect.

Approximate 80%-power minimum detectable effects for the models with both controls are 0.80 and 0.91 percentage points per SD, using (cluster-t critical + 0.842) times the clustered SE. This is a precision diagnostic, not observed power. Form-specific, added fixed-effect, two-way-cluster, ARKK-benchmark and active-word-list results are saved in the notebook. Holm adjustment covers only the four pooled both-control outcome tests.

Limits and next step

The 124-security universe is selected from 2026 holdings snapshots, not historical ARK ownership. Only 78 of 91 retained issuers appear in all five filing years. The number of formerly held firms missing from this snapshot cannot be established from the supplied data. Survivorship can distort trends as well as return tests; fixed effects do not remove this selection. Other limits are 20 aggregate quarters, full-corpus rather than live scoring, noisy daily event timing, multi-class valuation proxies, and overlapping post-filing windows. My highest-priority extension is a historical holdings universe with inclusion dates, followed by a genuinely held-out forecasting test.

Sources: Loughran and McDonald (2011), Journal of Finance 66(1), 35-65, equation (1); Fall 2026 assignment brief; instructor starter repository; LM dictionary documentation; Yahoo price-adjustment documentation; exchange_calendars; statsmodels. AI assistance is disclosed in AI_USE.md.